

NAME:P.Moshiya

REG NO:192572142

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy $< 90\%$ $\rightarrow$ system fails
- If latency > 70 ms $\rightarrow$ unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy $< 95\%$ $\rightarrow$ defects may be missed
- If processing time > 50 ms $\rightarrow$ production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your

decision using logical reasoning based on accuracy, processing time, and computational requirements.

1. Autonomous Vehicle Vision System Selection

Problem Analysis

An automobile company is developing a vision system for an autonomous vehicle. The system has three important constraints:

1. **Accuracy must be at least 90%.**
2. **Latency must not exceed 70 ms.**
3. **The vehicle has moderate computational resources.**

Therefore, a suitable method must satisfy **all three conditions simultaneously**. A method that fails even one condition cannot be selected.

Comparison of Methods

Method	Accuracy	Latency	Computational Requirement	Decision
Viola-Jones	82%	30 ms	Low	Rejected
YOLO	93%	60 ms	Moderate	Selected
Deep Learning	97%	90 ms	High	Rejected

1. Viola-Jones

Viola-Jones provides an accuracy of **82%** and a latency of only **30 ms**. Its low computational requirement is an advantage because it can operate with limited resources.

However, the required accuracy is **90%**, whereas Viola-Jones provides only **82%**.

Therefore:

- Accuracy condition: **Not satisfied**
- Latency condition: **Satisfied**
- Computation condition: **Satisfied**

Since accuracy is below the minimum requirement, Viola-Jones **cannot be selected** for this system.

2. YOLO

YOLO provides an accuracy of **93%** with a latency of **60 ms**. It requires moderate computation, which matches the computational resources available in the vehicle.

Therefore:

- Accuracy condition: **$93\% \geq 90\% \rightarrow$ Satisfied**

- Latency condition: **60 ms $\leq$ 70 ms $\rightarrow$ Satisfied**
- Computation condition: **Moderate $\rightarrow$ Satisfied**

YOLO satisfies **every constraint** given in the problem.

3. Deep Learning

Deep Learning provides the highest accuracy of **97%**, which is above the required 90%. However, its latency is **90 ms**, which is greater than the maximum acceptable latency of 70 ms.

It also requires **high computational resources**, whereas the vehicle supports only moderate computation.

Therefore:

- Accuracy condition: **Satisfied**
- Latency condition: **Not satisfied**
- Computation condition: **Not satisfied**

Hence, Deep Learning cannot be selected despite its higher accuracy.

Final Evaluation

YOLO is the **best and only feasible option** because it satisfies all the constraints simultaneously. Although Deep Learning has higher accuracy, its latency and computational requirements make it unsuitable. Viola-Jones is faster and computationally efficient, but its accuracy is insufficient.

Conclusion

YOLO should be selected for the autonomous vehicle vision system. It provides **93% accuracy**, **60 ms latency**, and requires **moderate computation**. Thus, it meets the accuracy, latency, and computational requirements and provides the most appropriate balance for the given application.

2. Smart Factory Product Inspection Selection

Problem Analysis

A manufacturing company wants to use computer vision to detect defective products on a production line. The system has three constraints:

1. **Accuracy must be at least 95%.**
2. **Processing time must not exceed 50 ms.**
3. **The available hardware supports moderate computation.**

The selected method must satisfy all these conditions because an inaccurate or slow system can negatively affect the production process.

Comparison of Methods

Method	Accuracy	Processing Time	Computational Requirement	Decision
Traditional Image Analysis	91%	25 ms	Low	Rejected
YOLO	96%	45 ms	Moderate	Selected
Deep Learning	98%	75 ms	High	Rejected

1. Traditional Image Analysis

Traditional Image Analysis has an accuracy of **91%** and a processing time of **25 ms**. It requires low computation, making it efficient.

However, the minimum required accuracy is **95%**. Since its accuracy is only 91%, it does not meet the accuracy requirement.

Therefore:

- Accuracy condition: **91% < 95% → Not satisfied**
- Processing time condition: **25 ms ≤ 50 ms → Satisfied**
- Computation condition: **Low → Satisfied**

Because accuracy is insufficient, Traditional Image Analysis cannot be selected.

2. YOLO

YOLO achieves **96% accuracy** and requires only **45 ms** processing time. It requires moderate computation, which matches the hardware available in the factory.

Therefore:

- Accuracy condition: **96% ≥ 95% → Satisfied**
- Processing time condition: **45 ms ≤ 50 ms → Satisfied**
- Computation condition: **Moderate → Satisfied**

YOLO satisfies **all three requirements**.

3. Deep Learning

Deep Learning achieves the highest accuracy of **98%**, which is excellent for detecting defective products.

However, its processing time is **75 ms**, which exceeds the maximum allowed 50 ms. It also requires **high computation**, while the available hardware supports only moderate computation.

Therefore:

- Accuracy condition: **Satisfied**

- Processing time condition: **Not satisfied**
- Computation condition: **Not satisfied**

Thus, Deep Learning is not suitable for the given real-time production environment.

Final Evaluation

YOLO is the only method that satisfies all the constraints. Traditional Image Analysis is faster and uses fewer computational resources, but its accuracy is insufficient. Deep Learning provides the highest accuracy but is too slow and computationally demanding for the given conditions.

Conclusion

YOLO should be selected for the smart factory product inspection system. It provides **96% accuracy**, **45 ms processing time**, and requires **moderate computation**. Therefore, it satisfies the accuracy, processing-time, and hardware requirements and is the most suitable method for real-time defect detection.

Overall Comparison

Application	Selected Method	Main Reason
Autonomous Vehicle Vision	YOLO	Meets 90% accuracy, 70 ms latency, and moderate computation
Smart Factory Inspection	YOLO	Meets 95% accuracy, 50 ms processing limit, and moderate computation

Overall conclusion: In both industry scenarios, **YOLO is the correct selection** because it is the only method that satisfies **all specified constraints simultaneously**.